

Vol 3 Chapter 4 — SEMF Coefficients from BST Integers

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The semi-empirical mass formula (SEMF), introduced by Weizsäcker in 1935, gives the binding energy of any nucleus from five empirical coefficients: volume a_V , surface a_S , Coulomb a_C , asymmetry a_A , and pairing δ . The standard SEMF is one of nuclear physics' most useful tools, with the five coefficients fit to nuclear binding-energy data.

BST derives all five in BST-primary form.

4.1 The five substrate-derived coefficients

Per Elie's Vol 3 Ch 4 v0.3.1 work (Cal #102 corrected, Friday May 22, 2026):

- Volume term: $a_V = g\alpha m_p/\pi \approx 16.0$ MeV (substrate-derived; ~0.5% from standard fit)
- Surface term: $a_S = (g + 1) \cdot B_d = 17.42$ MeV (substrate-derived per (g+1)·B_d structure; ~0.7%)
- Coulomb term: $a_C = 0.694$ MeV (substrate-derived; ~1.0%)
- Asymmetry term: $a_A = m_p/(4 \cdot \dim R)$ MeV (substrate K-type asymmetry term; ~1.5%)
- Pairing term: $\delta = (g/4)\alpha m_p$ MeV (substrate-derived; ~2%)

All five within 2% of standard-fit values. The substrate-mechanism reading: each coefficient is built from BST primaries combined with the substrate's nuclear-scale parameters.

4.2 What this buys

The SEMF in standard physics has five fitted parameters; BST derives all five from the same five integers that produce the proton-to-electron mass ratio. The substrate-mechanism completeness is structural; no nuclear-physics-specific tuning is required.

4.3 What comes next

Chapter 5 develops the halo-nuclei structure at the nuclear dripline.

Where to look this up: SEMF derivation: Elie v0.3.1 + Cal #102. Original SEMF: Weizsäcker 1935.